

10+ years experienced **Principal Software Engineer** building production GenAI, agentic AI, RAG, knowledge graph, recommendation, feature-serving, experimentation, fintech, blockchain, telecom, and enterprise platforms. Strong backend and data-platform depth across Python, Java, Scala, Spring Boot, FastAPI, Kafka, Redis, Aerospike, Databricks, LangChain, LangGraph, Google ADK, MCP, and distributed systems.

TECHNICAL SKILLS

AI / GenAI	Agentic AI, RAG, Knowledge Graphs, LangChain, LangGraph, Google ADK, MCP, Prompt Engineering
Programming	Python, Java, Scala, JavaScript
Frameworks	Spring Boot, FastAPI, Quarkus, REST APIs, Microservices, Event-Driven Design
Data Platforms	Kafka, Databricks, Feature Engineering, Data Pipelines, Recommendation Systems, Experimentation
Databases	MySQL, MongoDB, Redis, Aerospike, RocksDB, QdrantDB
DevOps	Git/GitLab, Jenkins, Docker, AWS

EXPERIENCE

- **MakeMyTrip**

Principal Software Engineer

 - Building production GenAI and agentic AI platform capabilities for travel and business workflows, including RAG, memory, tool orchestration, structured execution, evaluation, and observability.
 - Built **ContextOS**, a governed runtime and reliability harness for production AI agents with inspectable, replayable, policy-aware execution across orchestrator, cognitive core, typed contracts, and surface/BFF layers.
 - Reduced wrong-tool-call errors by **35%** using Tool-Call Reasonableness evaluation and added prompt-guideline enforcement to reduce workflow iteration loops before production use.
 - Reduced agentic workflow onboarding time through **Natural-Language**, allowing teams to define workflows through chat-driven skill onboarding and reusable workflow patterns.
 - Built GenAI marketing campaign automation that increased campaign throughput, enabled micro-cohort-based campaign generation, and reduced campaign creation cycle time from weeks to days.

Gurugram, IN

Feb 2025 – Present
- *Lead Software Engineer*

 - Led data-product and platform engineering across feature serving, experimentation, destination intelligence, and reusable customer-signal infrastructure.
 - Built internal feature platforms that process batch and real-time customer signals for data science, targeting, experimentation, and personalization workflows.
 - Optimized the Feature Broker service and reduced feature-serving latency from **15ms to 5ms**, a **66.7%** latency reduction and roughly 3x faster response time.
 - Developed experimentation and adaptive decisioning capabilities across audience definition, traffic allocation, exposure logging, reward computation, and treatment selection.
 - Built destination intelligence and knowledge graph foundations to organize travel context for discovery and AI-assisted experiences.

Jan 2023 – Feb 2025
- *Senior Software Engineer 2*

 - Built recommendation, personalization, shopper-signal, and feature-engineering systems for consumer travel products.
 - Developed trip-aware recommendation and Experience Engine capabilities that combine bookings, searches, preferences, trends, and intent signals for contextual cross-sell and travel discovery.
 - Improved targeting effectiveness using home-location-aware personalization, increasing CTR by **1.2%** for targeted users and contributing to approximately **100 incremental bookings per day**.
 - Built Kafka-driven data pipelines for shopper behavior, trip signals, user intent, personalization features, and downstream data-science systems.

Apr 2021 – Dec 2022
- *Senior Software Engineer 1*

 - Built backend services, campaign decisioning, rule-management, and internal configuration tooling for consumer travel workflows.

Oct 2019 – Mar 2021

- Built campaign rule, segmentation, and journey decisioning systems that converted customer attributes and business rules into configurable product and marketing decisions.
- Developed backend APIs and admin tooling for rules, experiments, configuration, and operational workflows across Java/Spring Boot services and Scala/Kafka/Redis workflows.

- **Broctagon Fintech Group** Noida, IN
Senior Software Engineer *Aug 2018 – Oct 2019*
 - Built fintech and blockchain operations platforms across trading CRM, wallet payments, token-sale workflows, KYC, SSO, and external integrations.
 - Improved backend throughput and reliability through Redis caching, asynchronous processing, optimized projections, secure APIs, and Spring Boot migration work.
- **Sopra Steria** Noida, IN
Software Engineer – Senior Software Engineer *Dec 2015 – Jul 2018*
 - Delivered enterprise telecom, media network-management, and retail ERP systems across alarms, dashboards, inventory, procurement, logistics, reporting, and operational workflows.
 - Built backend services, UI modules, reusable utilities, metadata-driven components, configurable data tables, and centralized configuration support for enterprise application teams.

PROJECTS

- **ContextOS: Governed Runtime for Production AI Agents** MakeMyTrip
AI Runtime Platform *Oct 2025 – Present*
 - Built governed AI-agent runtime across orchestrator, cognitive core, typed contracts, and surface/BFF layers; supports turn-first execution, intent classification, governed task graphs, HITL approvals, durable sessions, replayable DecisionRecords, capability governance, RAG, knowledge graphs, MCP tools, and schema-driven answers.
 - Reduced wrong-tool-call errors by **35%** using Tool-Call Reasonableness evaluation.
- **GenAI Marketing Campaign Automation Platform** MakeMyTrip
Marketing Automation *Sep 2025 – Present*
 - Built GenAI workflows that convert audience context, campaign requirements, and creative direction into targeted content for micro-cohort marketing. Increased campaign throughput and reduced campaign creation cycle time from weeks to days.
- **Agentic AI Framework and Memory Platform** MakeMyTrip
Agentic AI Platform *Mar 2025 – Sep 2025*
 - Built reusable autonomous-agent workflows combining memory, retrieval, tool orchestration, structured execution, and evaluation hooks. Reduced RAG pipeline onboarding from days to a few hours.
- **Vaani** Personal Project
Voice Productivity *2026*
 - Created a voice-first macOS productivity tool that reduces writing friction by turning spoken thoughts into clean, ready-to-use text directly inside the user's current workflow across email, docs, chat, tickets, code editors, and AI prompt fields.
- **LocalWiki** Personal Project
Second Brain *2026*
 - Built an LLM-maintained personal knowledge system inspired by Andrej Karpathy's Local Wiki guidelines, with immutable source ingestion, Obsidian-style compilation, source traceability, validation tooling, hybrid search, and multi-provider LLM maintainer workflows.
- **Travel Knowledge Graph and Destination Intelligence** MakeMyTrip
Destination Intelligence *Aug 2024 – Feb 2025*

- Built knowledge graph and RAG foundations for destination intelligence, enabling travel experiences to answer destination, itinerary, and discovery questions using structured facts, retrieved context, and LLM workflows.

- **Internal Feature Platform and Real-Time Feature Serving** MakeMyTrip
Feature Platform *Aug 2023 – Jul 2024*
 - Built feature infrastructure for batch and real-time customer signals used by recommendation, targeting, experimentation, and data science. Reduced Feature Broker latency from **15ms to 5ms**.
- **Experimentation and Adaptive Decisioning Platform** MakeMyTrip
Experimentation Platform *Jan 2023 – Sep 2023*
 - Built experimentation and decisioning capabilities including audience definition, traffic allocation, exposure logging, reward computation, and adaptive treatment selection.
- **Trip-Aware Recommendation and Experience Engine** MakeMyTrip
Personalization *Jan 2021 – Dec 2022*
 - Built recommendation systems that model customer journeys as itineraries and identify missing travel products across flights, hotels, cabs, buses, and related categories. Increased CTR by **1.2%** for targeted users and contributed to approximately **100 incremental bookings per day**.
- **Campaign Rule, Segmentation and Journey Decisioning** MakeMyTrip
Segmentation Based Experiences *Oct 2019 – Aug 2020*
 - Built rule-evaluation and campaign-decisioning systems that connect customer attributes, segments, journeys, experiments, and delivery controls using reusable rules instead of hard-coded release cycles.
- **ICO Fundraising, Wallet, and Bounty Management Platform** Broctagon
Blockchain Operations *Aug 2018 – Oct 2019*
 - Built blockchain CRM and operations modules for ICO fundraising, token-sale phase management, KYC, wallet payments, bonus distribution, bounty automation, and Ethereum smart-contract integration through Web3J.
- **Telecom and Media Network Management System** Sopra Steria
Network Management *Dec 2016 – Jul 2018*
 - Delivered backend and UI modules for telecom/media network management, including alarm, SLA, dashboard, and operational workflows for service-impacting event monitoring.

EDUCATION

- **Galgotias University** Greater Noida, IN
B.Tech in Computer Science and Engineering *Aug 2011 – Jul 2015*
CGPA: 8.61/10.0
- **Birla Vidya Mandir** Nainital, IN
Intermediate *2010*
Grade: 90.2%